

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Department of Mechatronics Engineering

Course Title : CAD Practice
Course Code : ME 2130
Experiment No : 01
Name of Experiment : Introduction to Basic Solid Modeling Techniques for 3D Part Creation
Date of Experiment : 4 July 2026
Date of Submission : 20 July 2026

Submitted By	Submitted To
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Experiment No: 01

Name of Experiment: Introduction to Basic Solid Modeling Techniques for 3D Part Creation

1.1 Objectives

1. To become familiar with the user interface and basic functionalities of SolidWorks.
2. To learn how to create and fully define 2D sketches using various sketching and dimensioning tools.
3. To understand and apply fundamental 3D modeling features, specifically the Extruded Boss/Base and Revolved Boss/Base tools.
4. To practice creating complete 3D solid parts from given 2D drawings by following a step-by-step modeling process.

1.2 Theory

solidworks ki, keno bebohar korbo, application kothay with ref.

1.3 Apparatus Required

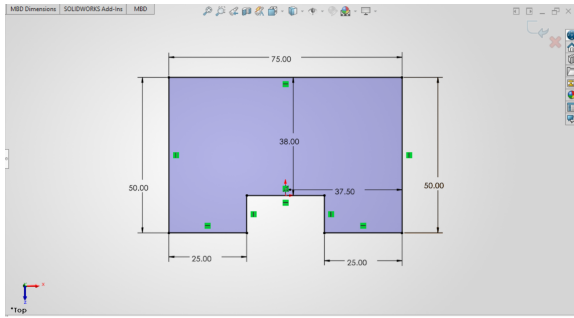
1. pc
2. solidworks software

1.4 Working

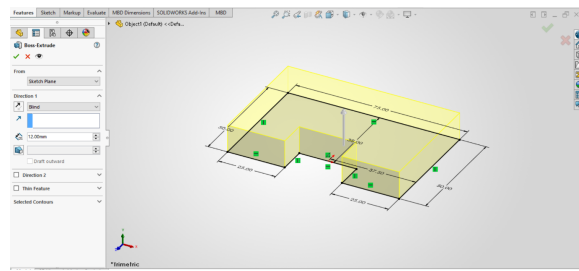
(example with screenshots) - unit selection - place selection - line drawing, dimensioning

1.5 Object 1

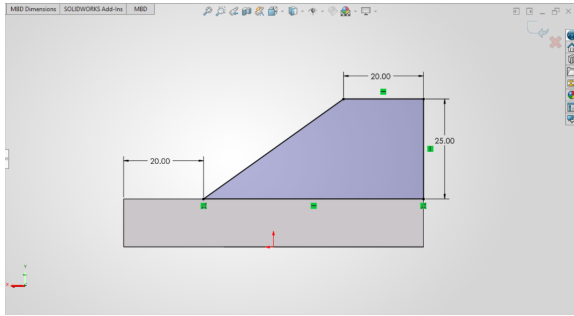
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna.



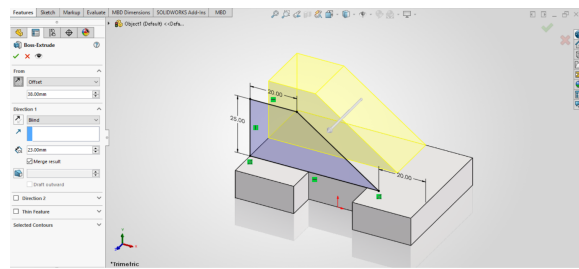
(a) 2D sketch of the base with dimensions



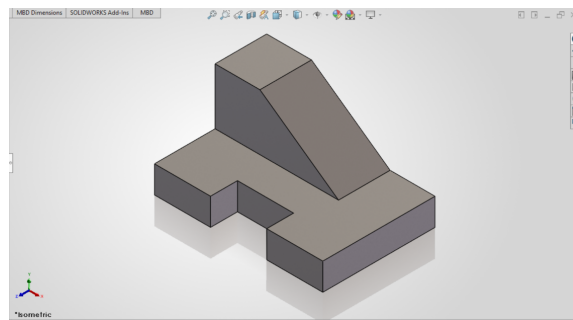
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

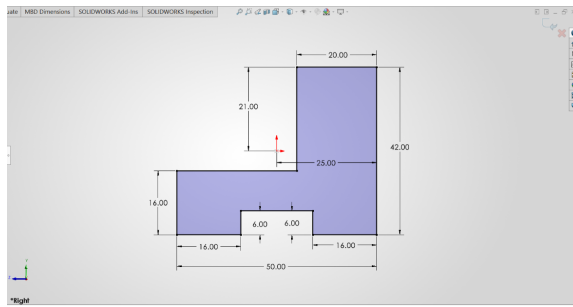


(e) Final Result

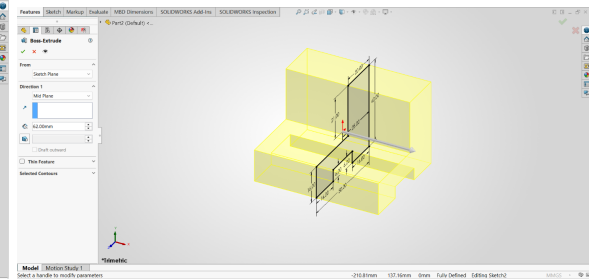
Figure 1.1: Snapshots while Designing Object 1 in Solidworks

1.6 Object 2

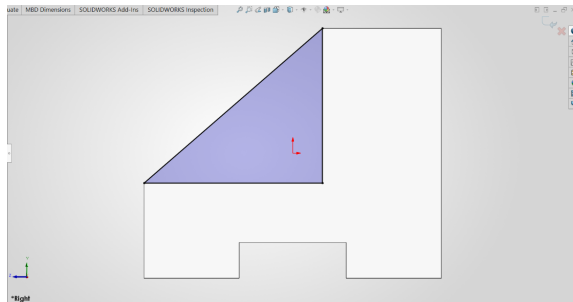
Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi.



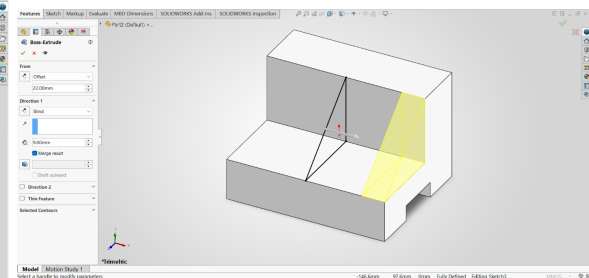
(a) 2D drawing of the right view with dimensions



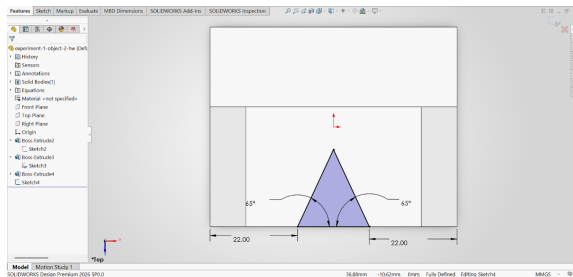
(b) Extruding from the midpoint



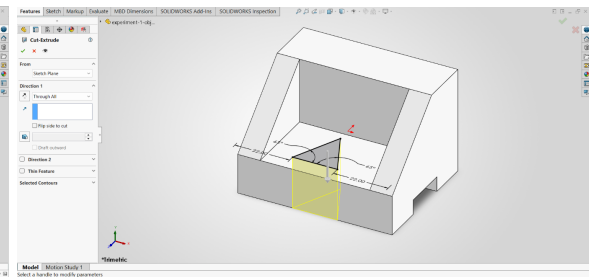
(c) 2D drawing of the extension on the right plane



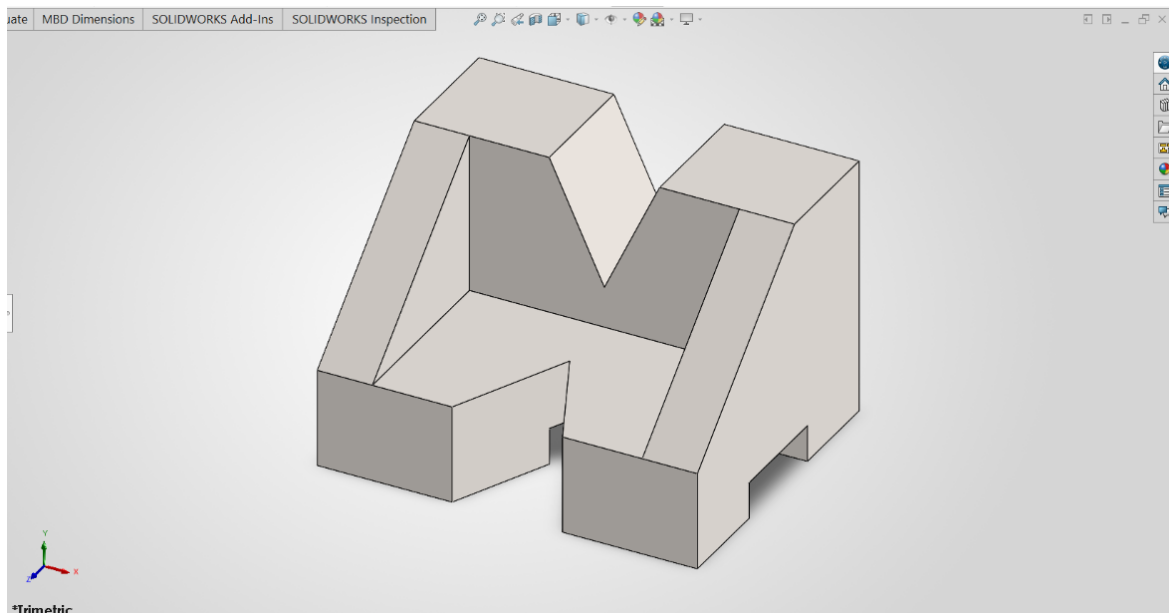
(d) Extruding the extension to the left



(e) 2D drawing of the cut-off portion on the bottom plane



(f) Using "Extruded Cut" feature on the portion to be cut-off



(g) Final result

Figure 1.2: Snapshots while Designing Object 2 in Solidworks

1.7 Object 3

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante.

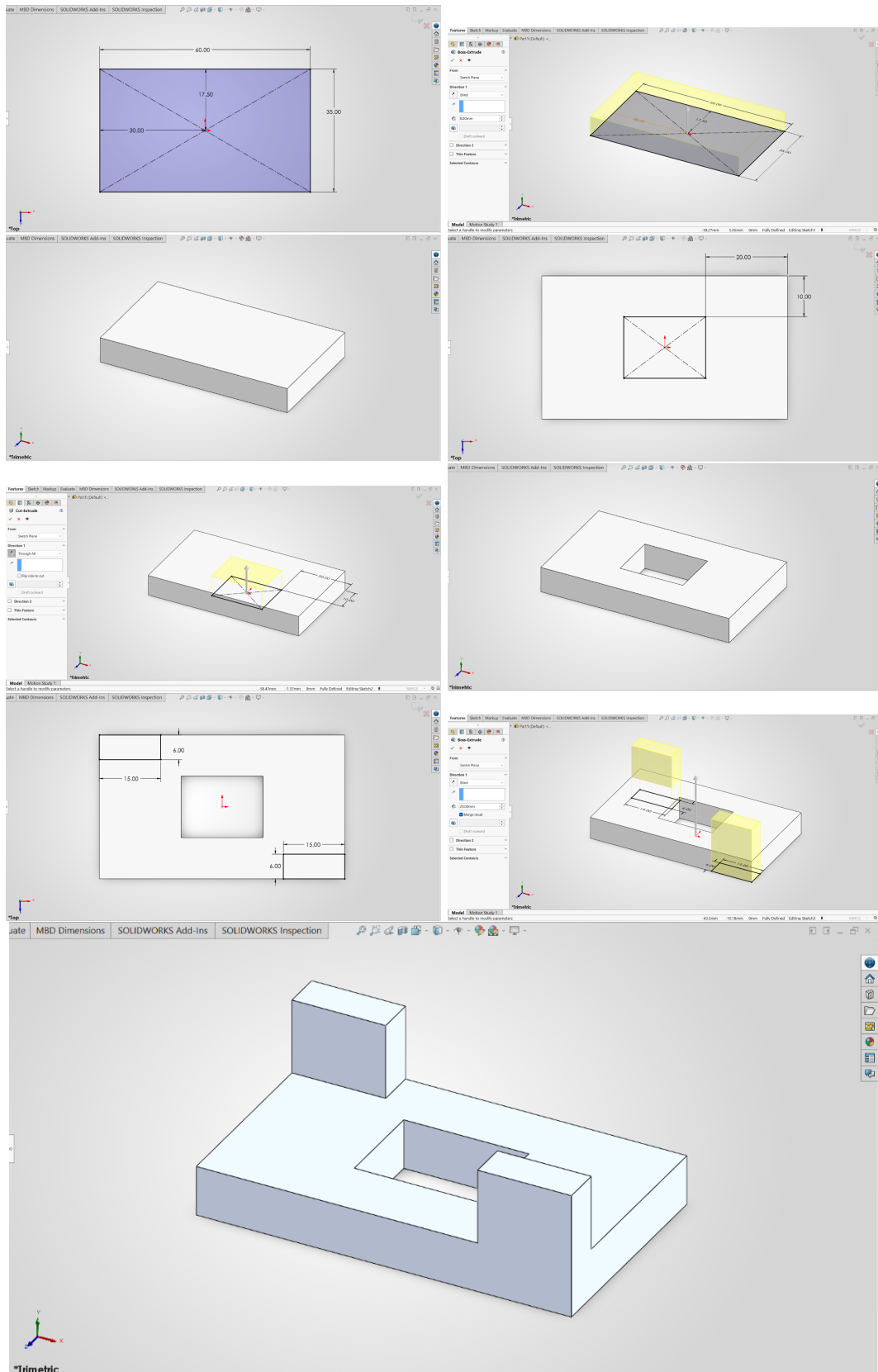


Figure 1.3: Snapshots while Designing Object 3 in Solidworks

1.8 Object 4

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

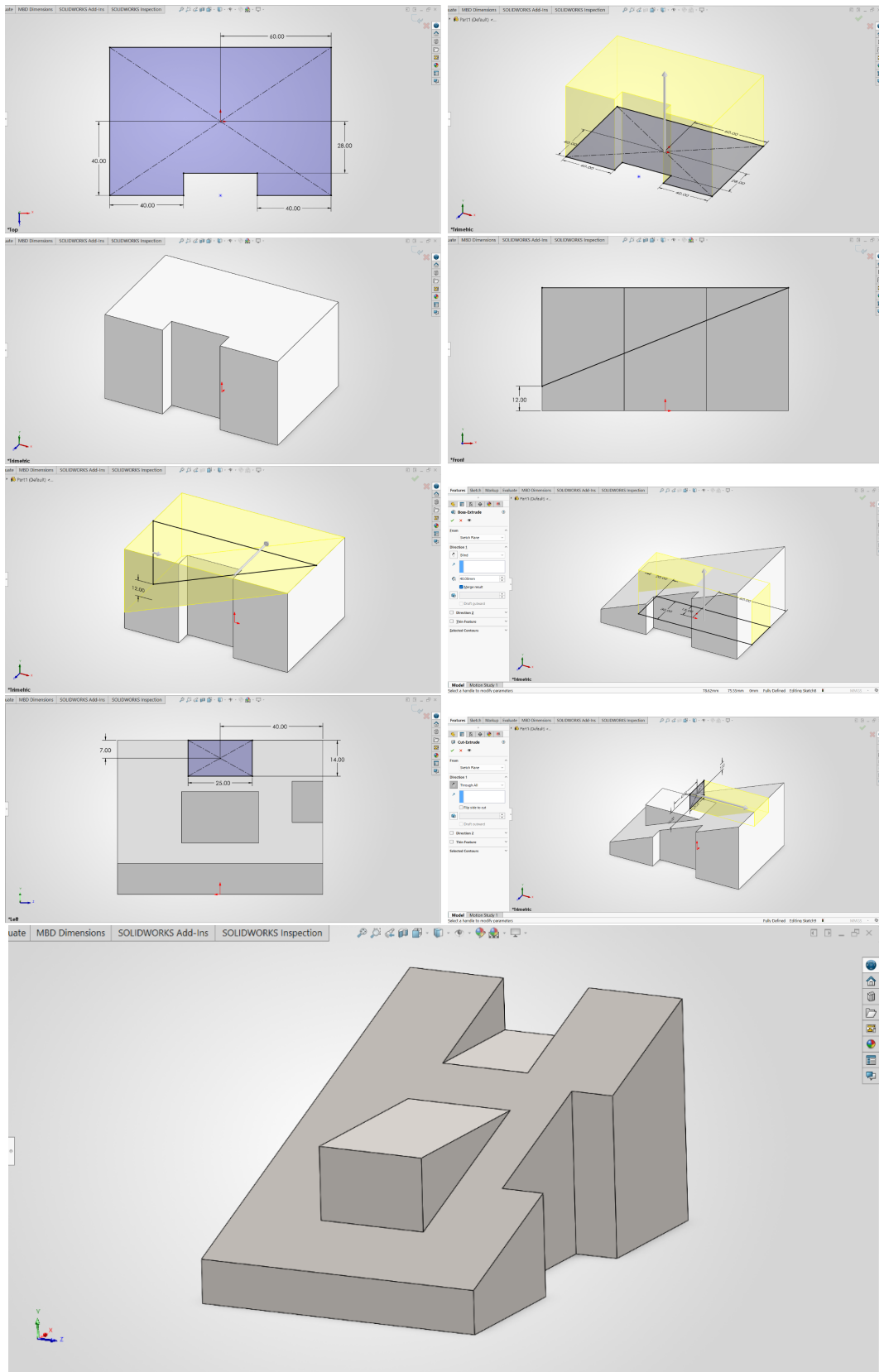
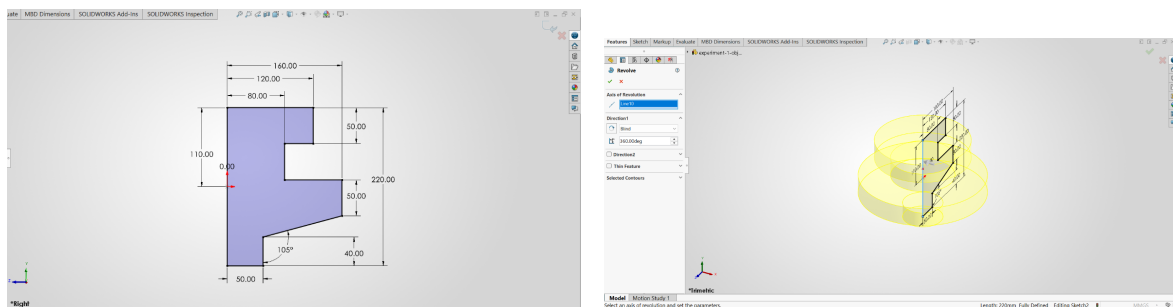


Figure 1.4: Snapshots while Designing Object 4 in Solidworks

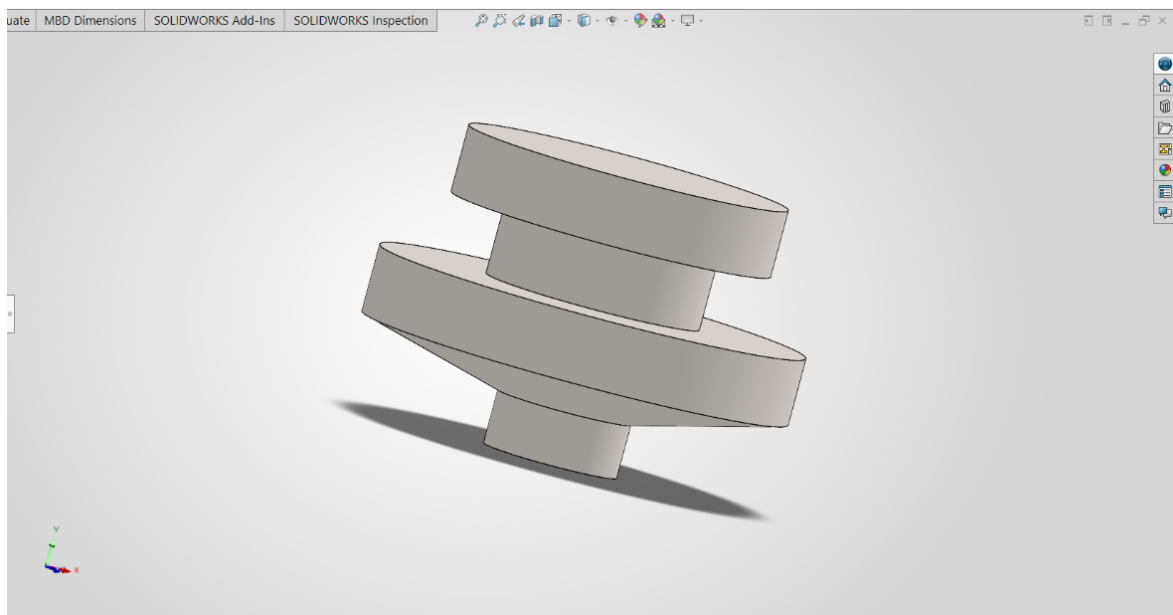
1.9 Object 5

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam.



(a) 2D drawing with dimensions

(b) Revolving around the vertical axis

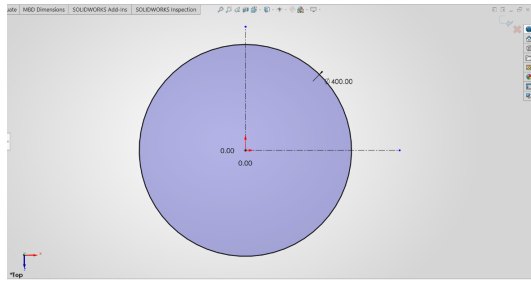


(c) Final Result

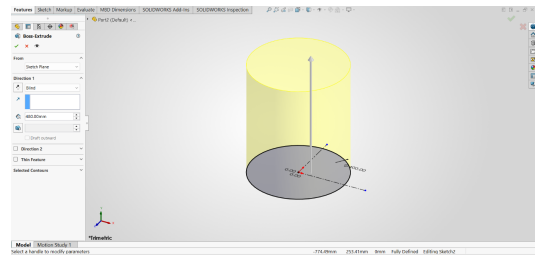
Figure 1.5: Snapshots while Designing Object 5 in Solidworks

1.10 Object 6

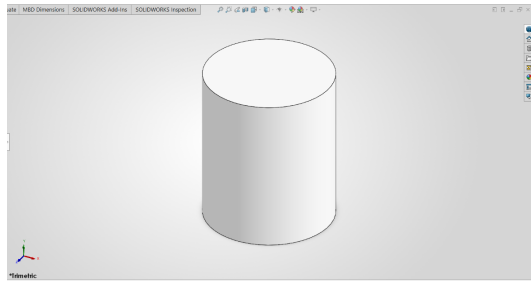
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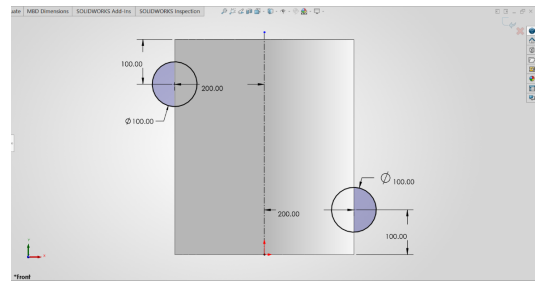
(a) 2D drawing with dimensions



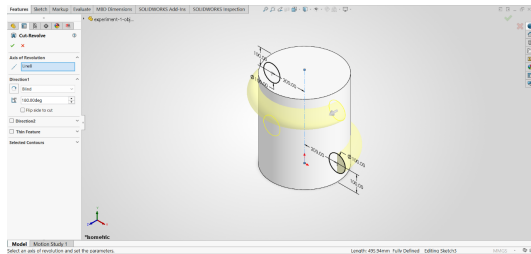
(b) Revolving around the vertical axis



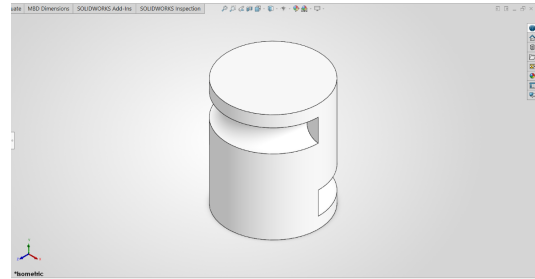
(c) Final Result



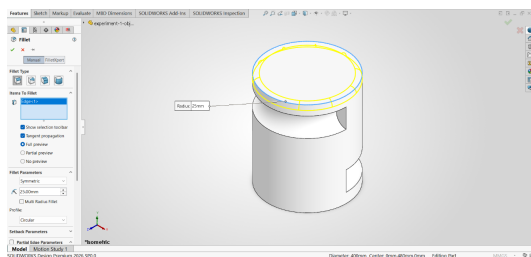
(d) 2D drawing with dimensions



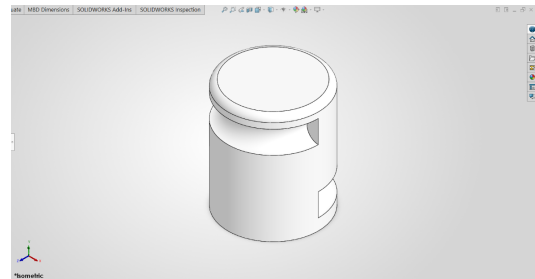
(e) Revolving around the vertical axis



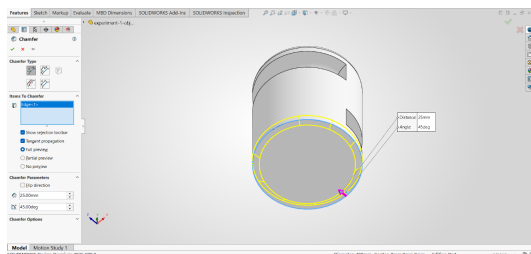
(f) Final Result



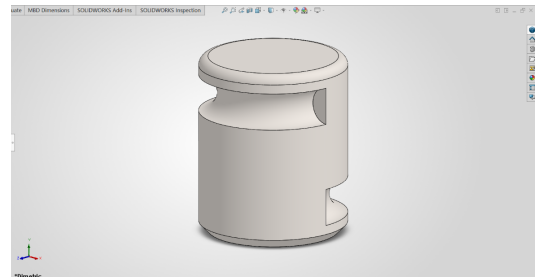
(g) 2D drawing with dimensions



(h) Revolving around the vertical axis



(i) Final Result

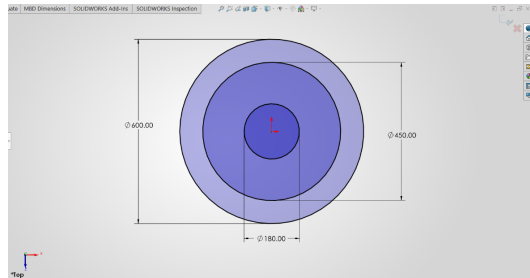


(j) 2D drawing with dimensions

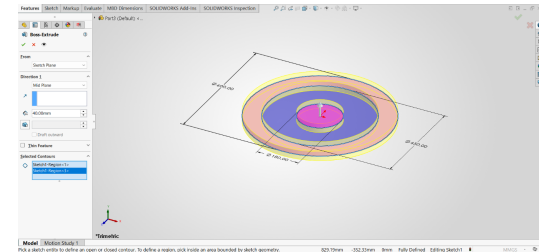
Figure 1.6: Snapshots while Designing Object 6 in Solidworks

1.11 Object 7

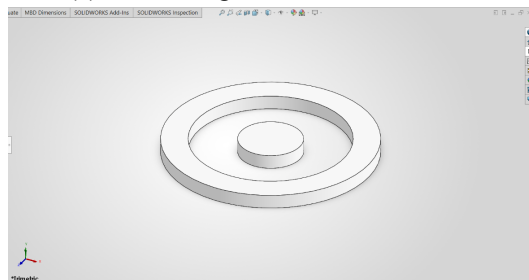
Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio.



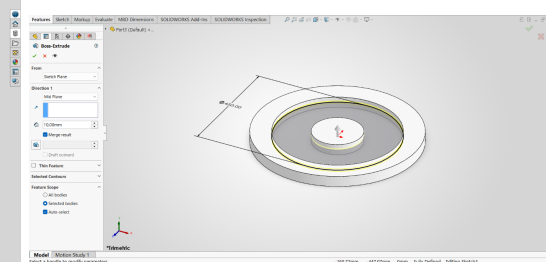
(a) 2D drawing with dimensions



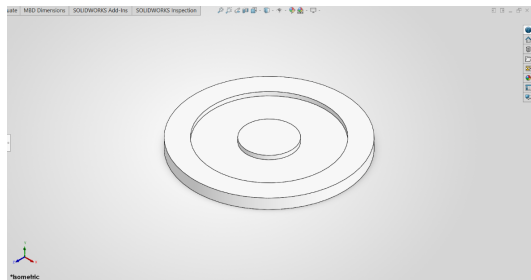
(b) Revolving around the vertical axis



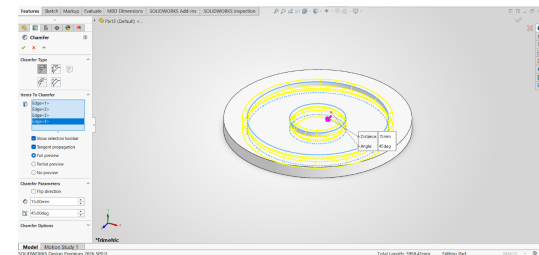
(c) Final Result



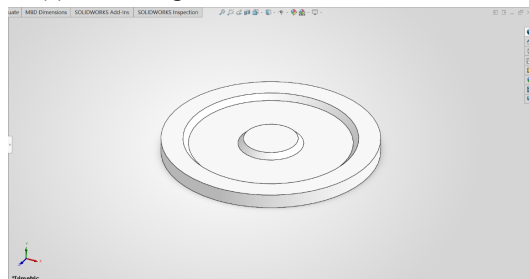
(d) 2D drawing with dimensions



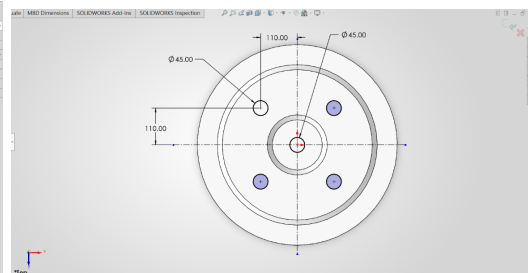
(e) Revolving around the vertical axis



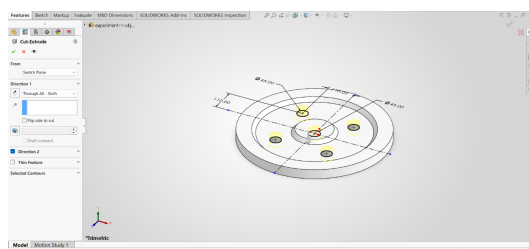
(f) Final Result



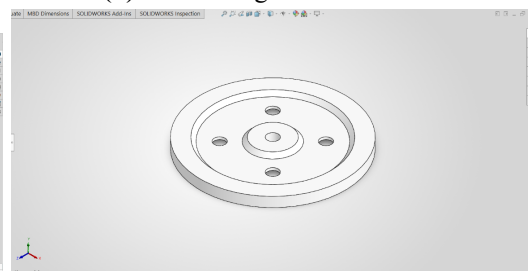
(g) 2D drawing with dimensions



(h) 2D drawing with dimensions



(i) 2D drawing with dimensions

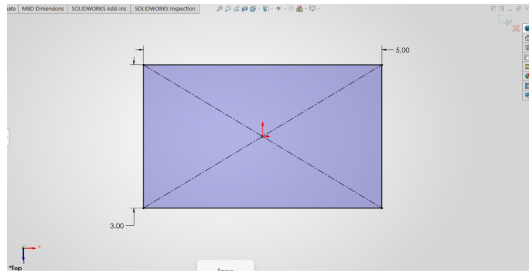


(j) 2D drawing with dimensions

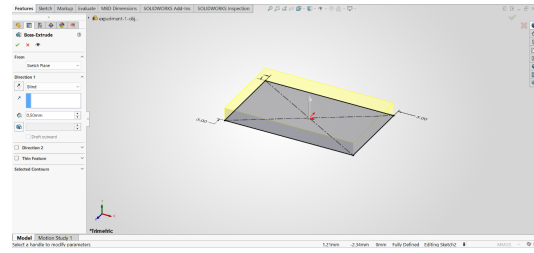
Figure 1.7: Snapshots while Designing Object 7 in Solidworks

1.12 Object 8

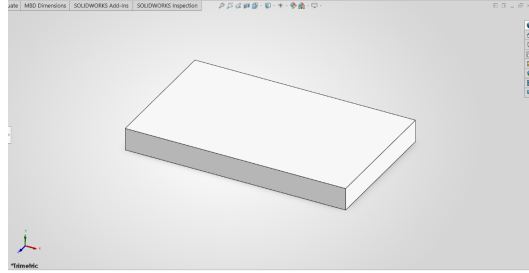
Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo.



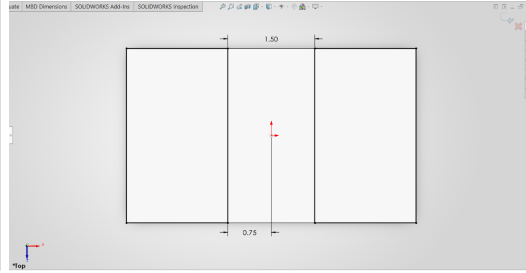
(a) 2D drawing with dimensions



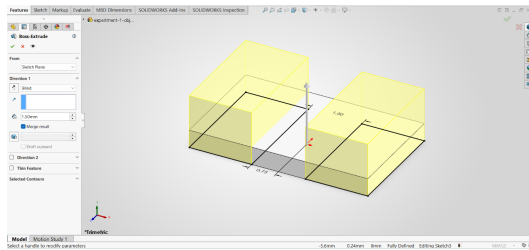
(b) Revolving around the vertical axis



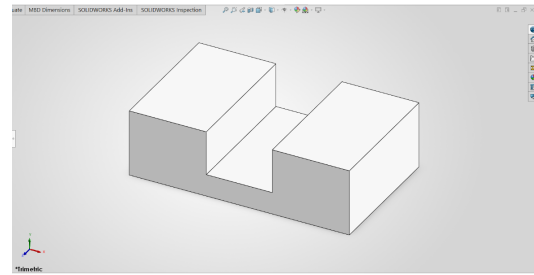
(c) Final Result



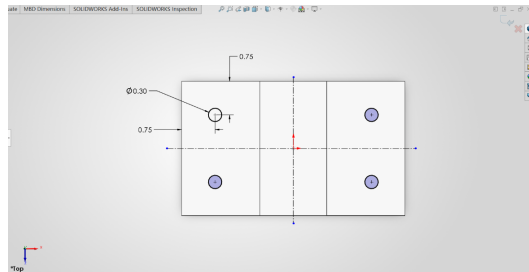
(d) 2D drawing with dimensions



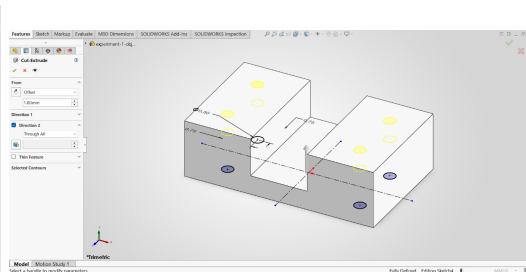
(e) Revolving around the vertical axis



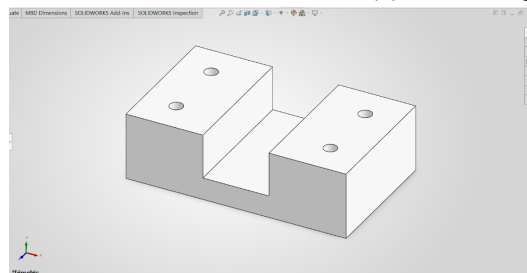
(f) Final Result



(g) 2D drawing with dimensions



(h) 2D drawing with dimensions

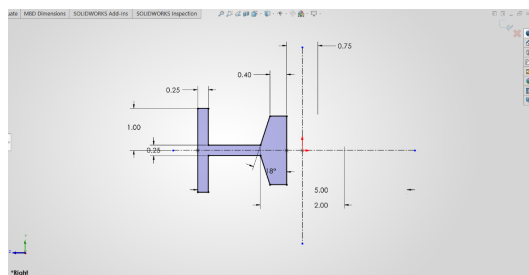


(i) 2D drawing with dimensions

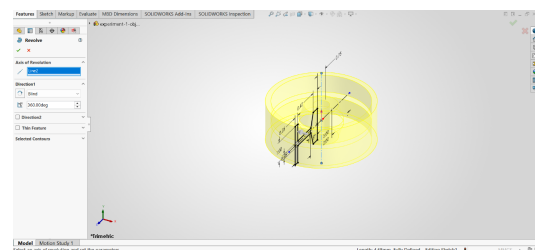
Figure 1.8: Snapshots while Designing Object 8 in Solidworks

1.13 Object 9

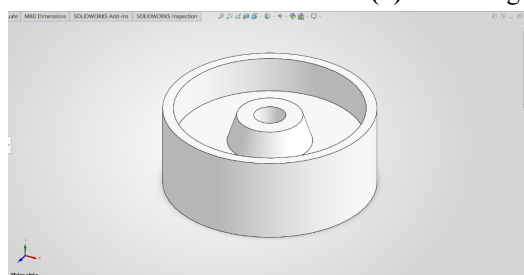
Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetur a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante.



(a) 2D drawing with dimensions



(b) Revolving around the vertical axis

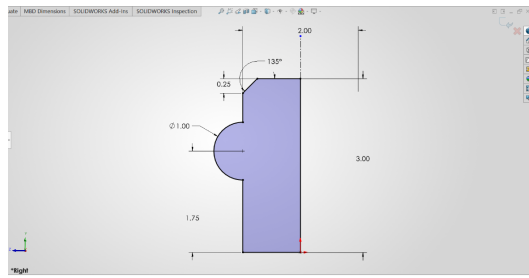


(c) Final Result

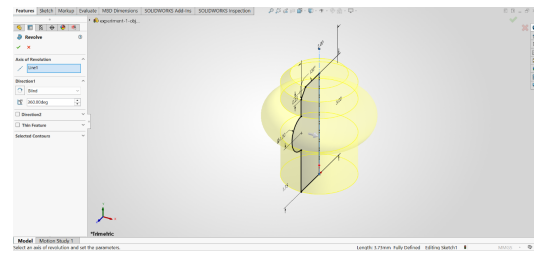
Figure 1.9: Snapshots while Designing Object 9 in Solidworks

1.14 Object 10

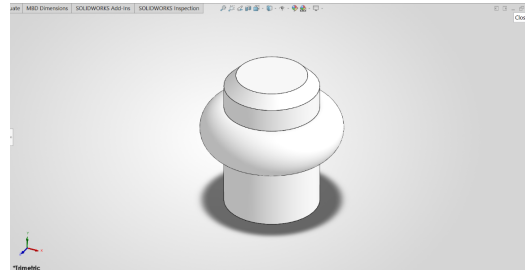
Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui.



(a) 2D sketch with dimensions



(b) Revolving the sketch around the vertical axis



(c) Final Result

Figure 1.10: Snapshots while Designing Object 10 in Solidworks

1.15 Discussion

1.16 Conclusion